

Coding with Kanak

Data Modeling Strategy & Multidimensional Models



Unit 2

Data Warehousing & Mining: Unit-2

Data Modeling Strategy

2.1 Data Modeling Strategy

A data modeling strategy defines how an organization will visually represent its data, outlining the entities, attributes, and relationships between them. It ensures functionality, accuracy, and security of a database system.

A strong data modeling strategy provides clarity, improves data quality and integrity, enhances performance, and ensures alignment with business objectives.

This strategy involves a step-by-step process, starting with identifying business requirements and creating a conceptual model, followed by developing detailed logical and physical models to guide database implementation and maintenance.

2.2 Steps in Implementing a Data Modeling Strategy

- Identify Business Requirements/Requirement Gathering:

-Understanding the purpose of the data model and what business problems it will solve.

-Involving stakeholders and domain experts to capture all information for the model.

- Create a Conceptual Model:

-Develop a high-level overview of the data, entities, and relationships involved.

-Create an abstract model of the data, focusing on the overall structure and relationships between the entities.

- Normalize the Model:

-Structure the data to reduce redundancy and improve integrity.

- It prevents anomalies and inconsistencies by minimizing data duplication.

- Develop a Logical Model:

- Create a detailed blueprint of the data structure, independent of a specific database.
- Develop a detailed model that defines entities, attributes, primary keys, and foreign keys.
- Details the data structure, ensuring relationships and constraints are clearly defined but without specifying the physical storage details.

- Design a Physical Model:

- Determine how the data will be stored and managed in the actual database system.
- Translate the logical model into a specific database system's structure.
- This model includes the actual tables, columns, data types, indexes, and other physical storage details needed for implementation.

- Implement the Model:

- Create the database and its associated structures based on the physical model.
- Implement the physical model into a database management system.

- Test and Refine:

- Run queries and reports to ensure accuracy and performance, making adjustments as needed.
- It validates the model to ensure it meets business requirements, performs well, and is accurate and complete.

- Document and Maintain:

- Provide clear documentation for the model and plan for updates as business needs evolve.
- Data models are living documents and should be updated as business needs evolve.
- Regularly test and refine the model to fix errors, improve performance, and adapt to new data and requirements

2.3 Fact Table and Dimension Table

Fact tables and dimension tables are key components of star or snowflake schemas . A fact table stores numeric, quantitative business event data ("facts"), while a dimension table provides descriptive, contextual details ("dimensions") that give context to these facts. A fact table contains foreign keys that reference the primary keys of related dimension tables, along with quantitative metrics.

Examples of a fact table include sales transactions, order histories or inventory changes — each record measures a business event. Examples of a dimension table include customer profiles, product catalogs and time/date hierarchies — each record adds context to a fact.

Characteristics of a Fact Table

- Contains quantitative data about the events or measurements of a business process.
- Holds foreign keys (the primary keys) of referenced dimension tables, plus its own primary key.
- Often one, central table of star schema or snowflake schema.
- Often the largest table in star schema or snowflake schema.

Characteristics of a Dimension Table

- Contains qualitative, descriptive data about the numerical data in a fact table.
- Holds its own primary key.
- Often multiple tables that surround the fact table in star schema or snowflake schema.
- Often smaller in size than the fact table in star schema or snowflake schema.

Difference Between Fact Table and Dimension Table

Fact Table Dimension Table

Fact table contains the measuring of the attributes of a dimension table.

Dimension table contains the attributes on that truth table calculates the metric.

In fact table, There is less attributes than dimension table.

While in dimension table, There is more attributes than fact table.

In fact table, There is more records than dimension table.

While in dimension table, There is less records than fact table.

Fact table forms a vertical table. While dimension table forms a horizontal table.

The attribute format of fact table is in numerical format and text format.

While the attribute format of dimension table is in text format.

It comes after dimension table. While it comes before fact table.

The number of fact table is less than dimension table in a schema.

While the number of dimension is more than fact table in a schema.

It is used for analysis purpose and decision making.

While the main task of dimension table is to store the information about a business and its process.

2.4 Star schema and other schemas

A schema is the blueprint that defines how data is organized, structured, and related within the warehouse, primarily using fact and dimension tables. It describes the logical arrangements of data.

Schema defines rules for how these data warehouses manage the names, descriptions, associated data items and aggregates within a data warehouse.

2.4.1 Schema Types

Different types of Schema describe the different logical arrangements of data.

- Star schemas

Star schemas are ideal for scenarios where simplicity and rapid query performance are the highest priorities, such as for dashboards and common business reports.

Structure:

A central fact table surrounded by multiple denormalized dimension tables, resembling a star.

Characteristics:

Uses fewer joins, leading to simpler and faster queries.

Advantages:

Easy to understand, design, and maintain, providing excellent query performance for analytical reporting.

Disadvantages:

High data redundancy due to denormalized dimensions, which can increase storage costs.

- Snowflake Schema

Snowflake schemas are more appropriate for very large and complex datasets where data integrity and storage efficiency are paramount, despite the added query complexity.

Structure:

A central fact table connected to normalized dimension tables that are further broken down into sub-dimension tables.

Characteristics:

Highly normalized structure, which reduces data redundancy but increases the complexity and number of joins needed for queries.

Advantages:

Improved data integrity, reduced storage costs, and better support for complex hierarchical relationships.

Disadvantages:

More complex queries and slower performance compared to star schemas.

- Other Schema Types
- Galaxy Schema:

A collection of multiple star and snowflake schemas, often used in larger data warehouses where different business processes are analyzed.

- Data Vault:

A hybrid approach that emphasizes agility and provides a flexible framework for data integration and warehousing, offering more adaptability than traditional star or snowflake schemas.

- Anchor Schema:

A lesser-known schema focused on defining a core anchor model for the data, which can then be populated with various satellite tables.

2.5 Multi dimensional data models

A Multidimensional Data Model is defined as a model that allows data to be organized and

viewed in multiple dimensions, such as product, time and location.

A multidimensional data model organizes data into "facts" (measurable values) and "dimensions" (contextual attributes like time, geography, or product), allowing for complex analytical queries and reporting, often visualized as a "data cube".

This structure facilitates rapid data analysis from multiple perspectives and making it ideal for Online Analytical Processing (OLAP) systems and decision -making processes in data warehouses.

2.6 Data Cube presentation of Fact Tables

A data cube presents information from a fact table by organizing its numerical measures (facts) and descriptive attributes (dimensions) into a multidimensional structure (Qube) for analysis.

This multidimensional view supports interactive analysis using operations like roll-up, drill-down, slice, and dice to explore data at different levels of detail across various perspectives such as time, product, and location.

Key Components of a Data Cube

- Fact Table:

The central component that stores quantitative measures (facts), such as sales amount or units sold, and foreign keys that link to the dimension tables.

- Dimensions:

Descriptive attributes or perspectives, like time, product, or customer that provide context for the facts. Dimensions are organized into hierarchies to allow for analysis at different granularities.

- Measures:

Numerical data within the fact table that can be aggregated, such as revenue or profit.

How Fact Tables are Presented in a Data Cube

- Multidimensional Model:

The fact table, along with its associated dimension tables, is transformed into a multidimensional structure, conceptually like a cube.

- Measures as Cells:

The numerical measures from the fact table are represented in the cells of the data cube.

- Dimensions as Axes:

The dimensions define the axes or perspectives along which the data is organized and viewed.

- Hierarchies and Cuboids:

Dimensions can have hierarchies (e.g., year > quarter > month), and the data cube is a lattice of cuboids, where each cuboid represents a different level of data summarization.

- Interactive Analysis:

Users can perform operations on the cube to gain insights:

Roll-up: Aggregating data to a higher level of the dimension hierarchy (e.g., summing sales from months to a year).

Drill-down: Moving from a high-level summary to more detailed data within a dimension (e.g., from a year to individual months).

Slice: Selecting a sub-cube by picking a single value for one of its dimensions (e.g., looking at sales for a specific month).

Dice: Selecting a sub-cube by specifying a range of values for multiple dimensions (e.g., sales of a product in a particular region during a specific quarter).

2.7 Data Warehouse Tools

A data warehouse tool is a software solution designed to collect, store, and analyze large volumes of structured and semi-structured data from various sources. These tools are essential for businesses to centralize data, improve analytics capabilities, and make informed decisions based on real-time and historical data insights.

When evaluating data warehouse tools, we need to consider the following key factors:

- Scalability: Ensure the tool can handle current data volume and scale as needed to grow.

Look for solutions that offer elastic scaling options, allowing us to adjust resources based on demand without significant downtime or performance issues.

- Performance: Look for fast query processing and data loading capabilities. Evaluate tools based on their ability to handle complex queries efficiently and support concurrent

users without compromising speed.

- **Ease of use:** Consider the user-friendliness of the interface. A good data warehouse tool should offer intuitive dashboards, drag-and-drop functionality for query building, and clear documentation to help users get up to speed quickly.
- **Integration capabilities:** Check compatibility with existing data sources and analytics tools. The ideal solution should offer a wide range of pre-built connectors and support for standard protocols to ensure seamless data flow across your entire data ecosystem.
- **Cost-effectiveness:** Analyze pricing models and total cost of ownership. Consider not just the upfront costs, but also ongoing expenses such as storage, compute resources etc.
- **Support and documentation:** Assess the quality of customer support and available resources. Look for vendors that offer multiple support channels, comprehensive documentation, active user communities, and regular training sessions or webinars.
- **Cloud vs. on-premises:** Determine which deployment model best suits organization's needs. Consider factors such as data residency requirements, existing infrastructure investments, and the level of control we need over our data warehouse environment.

2.8 Data Warehouse available Tools

1. Pelican.io

Pelican.io prioritizes user-friendliness and rapid deployment for data warehousing and business intelligence. Ideal for businesses seeking a user-friendly solution to get up and running quickly. Pelican is an all-in-one platform for all our data needs. It connects to all business applications, ETLs our data into a built-in data warehouse or Snowflake & BigQuery.

Features:

- **Unify Data:** Pelican seamlessly connects to over 100+ data sources.
- **Explore & Analyze:** Dive into data with a spreadsheet-like interface and Magical SQL.
- **Automate & Act:** Build data apps, set alerts, and share reports in minutes.

Considerations:

- May not be suitable for very large datasets (terabytes of data).
- Strong focus on using SQL and low-code Python, no built-in support for other scripting languages such as .NET or R

2. Snowflake

Snowflake offers a scalable, cloud -based data warehouse with el astic compute for on -demand processing power. It separates storage and compute, allowing for cost optimization . SQL query support makes data analysis accessible to existing users. Snowflake's ultra -scalable architecture adapts to our data volume needs, mak ing it a strong contender for organizations with growing datasets.

Features:

- Ultra-scalable architecture adapts to your data volume needs.
- Familiar SQL query support for data analysis by existing users.
- Cost-effective separation of storage and compute resources.

Considerations:

- Pricing structure can become complex for extensive deployments.
- Compute cost can increase rapidly.

3. Google BigQuery

Google BigQuery provides a cost -effective, serverless architecture with pay -per-use billing. It handles massive datasets with lightning -fast query speeds and boasts built -in machine learning for advanced data exploration. The serverless architecture eliminates infrastructure management needs, while the built-in machine learning capabilities empower us to uncover hidden patterns.

Features:

- Serverless architecture eliminates infrastructure management needs.
- Handles massive datasets efficiently with blazing-fast query speeds.
- Leverages built-in machine learning for advanced data exploration.

Considerations:

- Limited data transformation capabilities compared to some options.
- Security features might require additional configuration for specific compliance needs.

4. Microsoft Azure Synapse Analytics

Azure Synapse Analytics (formerly Azure Data Warehouse) is a cloud -native data warehouse integrated with other Azure services. It unifies data warehousing and big data analytics for comprehensive insights, offering visually interactive tools for user-friendly data exploration.

Features:

- Seamless integration with other Azure services for a unified data ecosystem.
- Unites data warehousing and big data analytics for broader data exploration.
- Offers visually interactive tools for user-friendly data exploration.

Considerations:

- Steeper learning curve compared to simpler tools due to its comprehensive nature.
- Pricing can vary depending on the Azure services used in conjunction.

5. Amazon Redshift

Amazon Redshift is a scalable data warehouse service built specifically for the AWS cloud environment. It's a cost-efficient option for analyzing large datasets stored in S3 and offers a familiar interface for AWS users. It scales efficiently to handle growing data volumes, making it a valuable option for organizations already invested in the AWS cloud.

Features:

- Cost-effective for analyzing large datasets stored in S3 buckets.
- Familiar interface for users comfortable with the broader AWS ecosystem.
- Scales efficiently to handle growing data volumes.

Considerations:

- May require additional configuration for optimal performance.
- Security features might require additional configuration for specific compliance needs.

6. Teradata

Teradata is an enterprise-grade data warehouse solution for mission-critical deployments. It offers robust security, high availability, and a scalable architecture for massive data volumes. Its robust security features making it a strong choice for organizations with sensitive data.

Features:

- Robust security features ensure data integrity and compliance.
- High availability architecture guarantees minimal downtime for critical operations.
- Massively scalable architecture handles enormous data volumes efficiently.

Considerations:

- Higher cost compared to some cloud-based options.
- Complex setup and management processes require significant IT expertise.

7. IBM Db2 Warehouse

Db2 Warehouse is a secure, reliable data warehouse built for integration with IBM's analytics ecosystem. It offers advanced data governance features and is designed for scalability and performance for demanding workloads. Db2 Warehouse integrates seamlessly with other IBM analytics tools, creating a unified environment for data management.

Features:

- Integrates seamlessly with other IBM analytics tools for a unified environment.
- Advanced data governance features ensure data accuracy and compliance.
- Scalable architecture handles high volumes of data and complex queries efficiently.

Considerations:

- May require familiarity with IBM technologies for optimal utilization.
- Potential vendor lock-in if heavily reliant on other IBM analytics services.

8. Oracle Autonomous Warehouse

Oracle Autonomous Warehouse offers self-driving data warehouse with automated management in the Oracle Cloud. It leverages machine learning for workload optimization and resource allocation, and integrates with other Oracle services. The self-driving architecture automates management tasks, simplifying data warehouse operations for organizations using the Oracle Cloud.

Features:

- Self-driving architecture automates management tasks for simplified operations.
- Leverages machine learning for workload optimization and resource allocation.
- Integrates seamlessly with other Oracle Cloud services for a unified data platform.

Considerations:

- Potential vendor lock-in if heavily reliant on other Oracle cloud services.
- Limited customization options compared to some open-source data warehouse solutions.

9. Cloudera

Cloudera is an open-source data platform offering a flexible and customizable data warehouse solution. It handles diverse data formats and sources but requires technical expertise for deployment and management. As an open-source platform, Cloudera provides greater flexibility and customization options compared to proprietary solutions.

Features:

- Open-source platform provides greater flexibility and customization options.
- Handles diverse data formats and sources for broader data integration.
- Cost-effective solution compared to some proprietary data warehouse options.

Considerations:

- Steeper learning curve compared to managed data warehouse services due to its open-source nature.
- Requires in-house technical expertise for deployment, configuration, and maintenance.

10. Amazon DynamoDB

Amazon DynamoDB is a NoSQL database service offering high performance and scalability for various data applications, including data warehousing. It's a good choice for real-time data workloads.

While not a traditional data warehouse solution, DynamoDB's flexibility and scalability make it suitable for organizations with real-time data streams that require warehousing alongside other functionalities.

Features:

- NoSQL database offers high scalability and performance for diverse data workloads.
- Flexible schema design adapts to evolving data models and requirements.
- Well-suited for real-time data applications with high data velocity.

Considerations:

- Not a traditional data warehouse solution; may require additional data transformation steps.
- Might not be ideal for complex data analysis due to its lack of built-in querying features.

12. PostgreSQL

PostgreSQL is a powerful, open-source relational database management system that also function as a data warehouse. It's a cost-effective option for organizations comfortable with open-source technologies. PostgreSQL offers a robust feature set for data management, querying, and security, making it a cost-effective alternative to traditional data warehouses for organizations with the in-house expertise to manage it.

Features:

- Open-source platform offers a cost-effective data warehousing solution.
- Robust feature set for data management, querying, and security.
- Large and active community provides extensive support and resources.

Considerations:

- Requires in-house expertise for setup, configuration, and ongoing maintenance.
- Limited scalability compared to some cloud-based data warehouse solutions.

13. MariaDB

MariaDB is another open -source relational database management system that can be used for data warehousing. It's a robust and secure option for organizations seeking a familiar and cost - effective solution, especially those already invested in the MySQL ecosystem. MariaDB provides a familiar SQL interface for users comfortable with relational databases, easing the learning curve for data management tasks.

Features:

- Open-source platform provides a cost-effective data warehousing solution.
- Familiar SQL interface for users comfortable with relational databases.
- High availability features ensure minimal downtime for critical operations.

Considerations:

- Requires in-house expertise for setup, configuration, and ongoing maintenance.
- Limited scalability compared to some cloud-based data warehouse solutions.

2.9 OLAP Models and Operations

OLAP is an acronym for On -Line Analytical Processing. OLAP is a software technology classification that allows analysts, managers, and executives to get insight into information through fast, consistent, interactive access to data that has been transformed from raw data to reflect the true dimensionality of the company as perceived by the clients.

Applications of OLAP

Here are several OLAP Analytics Operations:

- Roll-Up: Roll-Up is a technique for reducing dimensions by combining similar data into a single variable. For example, sales from several cities throughout the world can be

combined to form a country.

- Drill-Down: Drill -Down is the polar opposite of Roll -Up, which separates data for granular insights. For example, a financial institution's sales performance can be examined every quarter.
- Slice and Dice : Slice and Dice help with granular data analysis, but from a different angle. For example, you can examine a company's sales by different salespeople, which can then be evaluated further based on the things they sold.
- Pivot Table: Pivot Table s are used to summarise data and give an overall view of the relationship between two variables by rotating the axis of data. For example, using pivoting, salespeople can separate products for sale or vice versa.

Case study

For better understanding please follow the below data with the prescribed operations and corresponding output views.

OLAP operations:

There are five basic analytical operations that can be performed on an OLAP cube:

1. Drill down: In drill-down operation, the less detailed data is converted into highly detailed data.

It can be done by:

- o Moving down in the concept hierarchy
- o Adding a new dimension

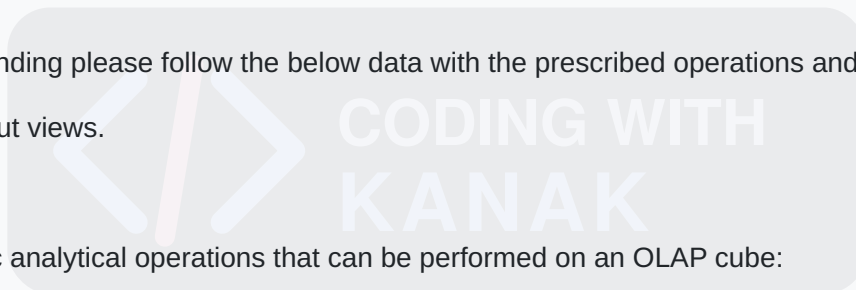
In the cube given in overview section, the drill down operation is performed by moving down in the concept hierarchy of Time dimension (Quarter -> Month).

2. Roll up: It is just opposite of the drill-down operation. It performs aggregation on the OLAP cube. It can be done by:

- o Climbing up in the concept hierarchy
- o Reducing the dimensions

In the cube given in the overview section, the roll-up operation is performed by climbing up in the concept hierarchy of Location dimension (City -> Country).

3. Dice: It selects a sub-cube from the OLAP cube by selecting two or more dimensions. In the cube given in the overview section, a sub-cube is selected by selecting following dimensions with criteria:



o Location = "Delhi" or "Kolkata"

o Time = "Q1" or "Q2"

o Item = "Car" or "Bus"

4. Slice: It selects a single dimension from the OLAP cube which results in a new sub-cube creation.

In the cube given in the overview section, Slice is performed on the dimension Time = "Q1".

5. Pivot: It is also known as rotation operation as it rotates the current view to get a new view of the representation. In the sub-cube obtained after the slice operation, performing pivot operation gives a new view of it.

2.10 Types of OLAP Models

These are the different types of OLAP Models:

- Relational OLAP Models (ROLAP)
- Multidimensional OLAP Models (MOLAP)
- Hybrid OLAP Models (HOLAP)

1. Relational OLAP Models (ROLAP)

1. ROLAP stands for Relational OLAP Model, a relational OLAP application.

2. These servers sit between a relational back-end server and the user front-end tools.

3. They save and manage warehouse data via a relational or extended -relational database management system, and they employ OLAP middleware to fill in the gaps.

4. ROLAP models execute optimization for each DBMS back end, aggregate navigation logic implementation, and other tools and services.

5. ROLAP systems mainly use data from relational databases, where the base data and dimension tables are stored as relational tables. This model also allows for multidimensional data analysis.

6. This method works by modifying the data in a relational database to simulate the slicing and dicing functionality of standard OLAP. Each method of slicing and dicing is essentially the same as adding a "WHERE" clause to a SQL statement.

Advantages

- Can Handle Large Volumes of Information: The data size limitation of ROLAP

technology depends on the data size of the underlying RDBMS. So, ROLAP itself does

not restrict the data amount.

Disadvantages

- Performance may be Slow: Each ROLAP report is a SQL query (or multiple SQL queries) in a relational database, and the query time can be extended if the underlying data is substantial.
- SQL Functions Limit ROLAP Technology: SQL statements are used to query a relational database, and SQL statements do not meet all needs.

2. Multidimensional OLAP Models (MOLAP)

1. MOLAP stands for Multidimensional OLAP Model, an application based on multidimensional DBMSs.
2. The foundation of a MOLAP model is a native logical model that allows multidimensional data and operations directly. Data is physically stored in multidimensional arrays and accessed using positional algorithms.
3. The scalability of ROLAP technology is generally higher than that of MOLAP technology.
4. One of the key differences between MOLAP and ROLAP is that data is summarised and stored efficiently in a multidimensional cube rather than a relational database. Data is formatted into proprietary forms per the client's reporting requirements in the MOLAP model, with computations pre-generated on the cubes.

Advantages

- Excellent Performance: MOLAP cubes are designed for quick data retrieval and are ideal for slicing and dicing activities.
- Can Conduct Sophisticated Calculations: When the cube is constructed, all evaluations are pre-generated. As a result, not only are complex calculations possible, but they also return rapidly.

Disadvantages

- It Can Only Handle a Certain Amount of Data: Because all calculations are done when the cube is produced, a vast quantity of data cannot be stored in the cube itself.
- Additional Investment is Required: Cube technology is usually proprietary and not already in use within the company. As a result, additional human and capital resources

will likely be required to implement MOLAP technology.

3. Hybrid OLAP Models (HOLAP)

1. HOLAP or Hybrid OLAP Model is an application that combines relational and multidimensional approaches.
2. HOLAP combines MOLAP and ROLAP's greatest characteristics into a single architecture. HOLAP systems store a larger amount of detailed data in relational tables, while aggregations are saved in pre-calculated cubes.
3. For defined data, HOLAP can drill down from the cube to the relational tables. A hybrid OLAP model is provided by Microsoft SQL Server 2000.

Advantages

- HOLAP combines the advantages of MOLAP and ROLAP.
- It allows quick access at all aggregate levels.
- HOLAP reduces disc space requirements by storing only the aggregate data on the OLAP models while keeping the detail records in the relational database. As a result, no duplicate copy of the detail record is kept.

Disadvantages

- HOLAP architecture is somewhat complex because it supports both MOLAP and ROLAP models.

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